

SAT UZBEKISTAN

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1.

A machine makes 8-inch, 5-inch, and 7-inch parts. During a certain day, the number of 8-inch parts that the machine makes is 3 times the number n of 5-inch parts, and the number of 7-inch parts is 12. During this day, the machine makes 100 parts total. Which equation represents this situation?

- A) $8(3n) + 5n + 7(12) = 100$
- B) $8n + 5n + 7n = 100$
- C) $3n + 12 = 100$
- D) $4n + 12 = 100$

2.

A linear model estimates the population of a city from 1991 to 2015. The model estimates the population was 59 thousands in 1991, 222 thousand in 2011, and x thousand in 2015. To the nearest whole number, what is the value of x ?

3.

For the linear function p , $p(c) = -2$, where c is a constant, $p(4) = 38$, and the slope of the graph of $y = p(x)$ in the xy -plane is 8. For the linear function t , $t(c) = -3$ and $t(5) = 51$. What is the slope of the graph of $y = t(x)$ in the xy -plane?

4.

$$f(x) = 3(x - a)(x - b)(x - c)$$

The function f is defined by the given equation, where a , b , and c are distinct constants. When $a < x < b$, the value of $f(x)$ is positive. The graph of $y = f(x)$ in the xy -plane contains the point (r, s) , where r and s are constants. If $s = 8$, which of the following could be true?

- I. $r < a$
 - II. $b < r < c$
 - III. $r > c$
- A) I only
 B) III only
 C) I and III only
 D) II and III only

5.

$$26z^{18} + bz^9 + 70$$

In the given expression, b is a positive integer. If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

6.

In the xy -plane, the graph of function f , where $y = f(x)$, has exactly 8 x -intercepts. One of these x -intercepts is $(17, 0)$. The rational function g is defined by

$$g(x) = \frac{f(x)}{x - 17}.$$

In the xy -plane, how many x -intercepts does the graph of $y = g(x)$ have?

7.

$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 6 . What is the value of a ?

- A) -84 C) 3
 B) -3 D) 84

8.

An object is launched from a height of 121 feet above the ground. A quadratic function f models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 2 seconds after the object is launched, it reaches a maximum height of 185 feet above the ground. Which equation defines f ?

- A) $f(t) = -16(t + 185)^2 - 2$
- B) $f(t) = -16(t - 185)^2 + 2$
- C) $f(t) = -16(t + 2)^2 - 185$
- D) $f(t) = -16(t - 2)^2 + 185$

9.

The function h is defined by

$$h(x) = -\sqrt{x^2 + bx + c},$$

where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{334})$. If $h(m) = 0$, what is the greatest possible value of m ?



10.

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(32) > 0$, $f(35) < 0$, and $f(38) > 0$, which of the following could be the value of $a + b$?

- A) 35
- B) 37
- C) 67
- D) 69

11.

The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	27%
18–40 years old	24%
41–65 years old	28%
Greater than 65 years old	21%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?

- A) 0.21
- B) 0.29
- C) 0.35
- D) 0.41

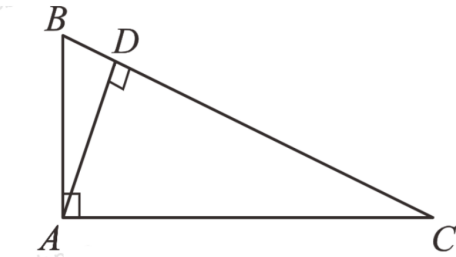
12.

The table summarizes the distribution of age and assigned group for participants in a study.

	0-9 years	10-19 years	20+ years	Total
Group A	14	19	27	60
Group B	27	7	26	60
Group C	19	34	7	60
Total	60	60	60	180

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age? (Express your answer as a decimal or fraction, not as a percent.)

13.



In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . The measure of $\angle ABD$ is w° , the measure of $\angle CAD$ is z° , and $AD = 178$. If $\cos w^\circ = \frac{9}{41}$, what is the value of $\tan z^\circ$?

14.

A right square pyramid has a total surface area of 70,560 square inches, and the combined surface area of the four lateral faces of this pyramid is 38,160 square inches. What is the height, in inches, of this pyramid?

15.

In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment WX is perpendicular to segment YZ . The length of segment YW is 612, and the length of segment WX is 459. What is the value of $\tan Z$?

16.

In the xy -plane, line k and line l are perpendicular and intersect at the point $(2, 8)$. If line k is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line l ?

☐ (A) $\left(3, 8 - \frac{1}{m}\right)$

☐ (B) $\left(3, 8 + \frac{1}{m}\right)$

☐ (C) $(3, 8 - m)$

☐ (D) $(3, 8 + m)$

17.

A polygonal chain is a connected series of line segments whose length is defined as the sum of the lengths of the line segments it is made of. A polygonal chain with a length of 33 centimeters (cm) is extended by adding line segments that each have a length of $8 - r$ cm. Which function f represents the total length, in cm, of the extended polygonal chain after x line segments are added?

☐ (A) $f(x) = 33(8 - r) + x$

☐ (B) $f(x) = 8x - rx + 33$

☐ (C) $f(x) = 8x - r + 33$

☐ (D) $f(x) = (41 - r)x$

18.

Jackson mows lawns in the summer. When Jackson mows a lawn with a fence he charges \$1.03 per minute. When Jackson mows a lawn without a fence, he charges \$0.56 per minute. Jackson spends a total of 15 hours mowing lawns with and without a fence in a week and charges a total of \$701.40 from mowing lawns during that week. How many minutes does Jackson spend mowing lawns with a fence during that week?

19.

$$2x^2 + 2y^2 + tx + ty - 78 = 0$$

The given equation, where t is a positive constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{41}$

What is the value of t ?

20.

The rectangular region of land is divided into 40 square lots of equal area R , in square units. The length of the region is 1.6 times the width of the region. If the width of the region is $x\sqrt{R}$ units, what is the value of x ?

21.

$$y = 2x^2 - 19x + 40$$

$$y = x + a$$

In the given system of equations, a is a constant. The graph of the equations in the given system intersect at exactly one point (x, y) , in the xy -plane. What is the value of x ?

- | | |
|--------|-------|
| A) -10 | C) 5 |
| B) -5 | D) 10 |

22.

Arturo builds and sells two types of steel tables. The revenue Arturo generates from selling each square table is 2 times his cost to build a square table, and the revenue Arturo generates from selling each hexagonal table is 2.5 times his cost to build a hexagonal table. In one year, Arturo's total cost to build tables was \$25,740, and he generated a total revenue of \$61,875 from selling all the tables he built that year. For that year, how much greater is the revenue Arturo generated from selling hexagonal tables than the revenue he generated from selling square tables?